

ALGEBRAIC STRUCTURE
Sub Group and examples

Lecture—3-4

ALGEBRAIC STRUCTURE



Order of an element

➤ The order of an element a in G is the smallest positive integers n such that $a^n = e$.

If no such integer exists, we say a is of infinite order. The order of an element a is denoted by $o(a)$

➤ Example: let $G = \{1, -1, i, -i\}$ be a commutative group under the multiplication. Find the order of every element.

$$G = \{1, -1, i, -i\}, X^*$$

$$\underline{a_1} = \underline{e} \quad \checkmark$$

Solution:

$$e = 1$$

$$\cancel{a_1} (1)' = 1 \quad ; \quad o(1) = 1$$

$$(-1)^2 = 1 \quad ; \quad o(-1) = 2$$

$$(i)^4 = 1 \quad o(i) = 4$$

$$(-i)^4 = 1 \quad o(-i) = 4$$

ex:- $G = \{1, -1, i, -i\}$ $\times$

find the order of each element.

Soln:- The identity element of this group is 1.

$$o(1) = 1 \quad (1)^1 = 1 \quad o(-i) = 4$$

$$o(-1) = 2 \quad (-1)^2 = 1$$

$$o(i) = 4 \quad ; \quad (i)^4 = 1 \quad (-i)^4 = 1$$

Order of an element

➤ Solution: Let $G = \{1, -1, i, -i\}$ the order of

$$O(1) = 1$$

$$O(-1) = -1 \times -1 = (-1)^2 = 1 \quad \therefore O(-1) = 2$$

$$O(i) = i \times i \times i \times i = (i)^4 = 1 \quad \therefore O(i) = 4$$

$$O(-i) = -i \times -i \times -i \times -i = (-i)^4 = 1$$

$$\therefore O(-i) = 4$$

Order of an element

➤ Find the order of every element in the multiplicative group $G = \{a, a^2, a^3, a^5, a^6 = e\}$

Solution: The identity element of the group is $a^6 = e$

$$O(a) = a \times a \times a \times a \times a \times a = a^6 = e \quad \therefore O(a) = 6$$

$$O(a^2) = a^2 \times a^2 \times a^2 = a^6 = e \quad \therefore O(a^2) = 3$$

➤ Find the order of every element in the multiplicative group $G = \{a, a^2, a^3, a^5, a^6 = e\}$

Soln: $G = \{a, a^2, a^3, a^5, a^6\}$
 $a^6 = e$, operator $\rightarrow \times$
 ~~a^6~~ $a^6 = e$ $O(a) = 6$
 $(a^2)^3 = a^6 = e$ $O(a^2) = 3$
 $(a^3)^2 = a^6 = e$ $O(a^3) = 2$
 $(a^5)^6 = (a^6)^5 = e$ $O(a^5) = 6$

➤ Find the order of every element in the multiplicative group $G = \{a, a^2, a^3, a^5, a^6 = e\}$

Soln.:- The identity element of this group is $a^6 = e$

$$(a)^6 = e, \quad o(a) = 6$$

$$(a^2)^3 = a^6 = e, \quad o(a^2) = 3$$

$$(a^3)^2 = a^6 = e, \quad o(a^3) = 2$$

$$(a^5)^6 = (a^6)^5 = e, \quad o(a^5) = 6$$

Order of an element

$$O(a^3) = a^3 \times a^3 = a^6 = e \quad \therefore O(a^3) = 2$$

$$O(a^5) = a^5 \times a^5 \times a^5 \times a^5 \times a^5 \times a^5 = a^{30} = (a^5)^6 = e \quad \therefore O(a^5) = 6$$

$$O(a^6) = a^6 = e \quad \therefore O(a^6) = 1$$

- Sub Group

Let $(G,*)$ be a group and H is a subset of G . $(H,*)$ is said to be subgroup of G if $(H,*)$ is also group.

Example: if $(G,*) = (\mathbb{Z}, +)$
and $(H,*) = (\text{Even integer}, +)$ then $(H,*)$ is
subgroup of $(G,*)$.

Example: if $(G,*) = (\mathbb{Z}, +)$
and $(H,*) = (\text{odd integer}, +)$ then $(H,*)$ is not
subgroup of $(G,*)$.

Sub group :- (i) $(G, *) \rightarrow$ group

A set H is a subgroup of G
(i) if

$$(i) \quad H \subseteq G$$

(ii) $(H, *) \rightarrow$ group

for ex:- ~~$(G, +)$~~ $(\mathbb{Z}, +)$
is group

Let H is even numbers

Test, H is a subgroup or not.

(i) $\underline{H} \subseteq \mathbb{Z}$ (yes)

$$H = \{-6, -4, -2, 0, 2, 4, 6, \dots\}$$

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, 4, \dots\}$$

$$H \subseteq \mathbb{Z} \quad \checkmark$$

(ii) H is a group or not

H is group.

(i) Closure, ✓

(ii) Associative ✓

(iii) Identity element ✓

(iv) Inverse ✓

∴ H is a group under addition.

∴ H is a subgroup.

ex: $\text{Test } (H, +)$ is a subgroup
or not of $(G, +) = (\mathbb{Z}, +) \Rightarrow$
 $(H, +) = (\text{odd integers}, +)$

Solution: $H = \{-5, -1, 0, 1, 3, 5, \dots\}$

(1) closure law; $\times$

$\therefore \Rightarrow$ It is not a group
 $\Rightarrow$ It is not a subgroup of G .

Subgroup :- $(G, *) \rightarrow \text{Group}$

$H \subseteq G$ $(H, *) \rightarrow \text{if this is a group}$

$(H, *)$ is a subgroup.

(i) H is subset of G

(ii) H is group under same binary operation of G .

ex:- $G = (\mathbb{Z}, +)$

$$H = (\text{even integers}, +)$$

(I) Closure $\checkmark$

(II) Associative law $\checkmark$

(III) 0 $\checkmark$

(IV) ~~every~~ inverse is $\checkmark$

$\therefore H$ is group, $H \subset G$
imp.

$$G(\bar{E}, +), \quad H = (\text{odd integers})_+$$

(1) Close $\times$

$$3, 5, \quad 3 + 5 = \underline{\underline{8}}$$

$\therefore H$ is not a sub group.

- Sub Group

Example: the multiplicative group $\{1, -1\}$ is a subgroup of the multiplicative group $\{1, -1, i, -i\}$

NOTE:

$(G, *)$ and $(e, *)$ where e is the identity element are called Improper or trivial subgroup

- Properties of Sub Group:

Theorem: The identity element of a subgroup is the same as that of the group.

Proof: let H is a subgroup of G .

$$\therefore H \subseteq G$$

$$\text{If } a \in H \Rightarrow a \in G$$

Let e_1 and e_2 are two identity element of H and G

$$\therefore a \cdot e_1 = a = a \cdot e_2$$

$$\Rightarrow a \cdot e_1 = a \cdot e_2$$

$$e_1 = e_2$$

$$e_1 = a \cdot e_1 = a \cdot e_2 = e_2$$

Theorem: A non empty subset H of a group G is a subgroup of G if and only if

$$(i) \quad a \in H, b \in H \Rightarrow a * b \in H$$

$$(ii) \quad a \in H \Rightarrow a^{-1} \in H$$

Necessary condition : Since H is a subgroup $\Rightarrow$
 $(i), (ii)$ hold.

Sufficient Condition: if $(i), (ii)$ satisfies we have to show that H is a subgroup.

Theorem: A non empty subset H of a group G is a subgroup of G if and only if

(i) $a \in H, b \in H \Rightarrow a * b \in H$

(ii) $a \in H \Rightarrow a^{-1} \in H$

Proof $\div$ ~~$H \subseteq G$~~ ^{Given} $H \subseteq G$, H is non empty

Q.1 H is a subgroup of G if

$\checkmark$ (i), (ii)

Q.2 If (i) & (ii) exist then
 H is a subgroup.

(i) ~~is~~ closure $\checkmark$ (ii) $\checkmark$ $H \subseteq G$

(ii) $a \in H, a^{-1} \in H \Rightarrow a * a^{-1} = e \in H$

$$\begin{array}{ll}
 \text{(11)} & \forall a, b \in H, \quad \underline{\underline{a \times b \in H}} \\
 \text{(12)} & \forall a \in H, \quad \underline{\underline{a^{-1} \in H}} \\
 & b \in H, \quad \underline{\underline{b^{-1} \in H}}
 \end{array}
 \quad \left. \vphantom{\begin{array}{l} \text{(11)} \\ \text{(12)} \end{array}} \right\}$$

$$a \times a^{-1} \in H \quad (i)$$

$$e \in H$$

$$\underline{\underline{(13)}} \quad \underline{\underline{H \subseteq G}}$$

- Proof of theorem conti.....

As the element of H is also the element of G and the element of G satisfy the association law for the binary operation. Therefore the element of H will also satisfy the associative law.

$$\begin{aligned}\text{Now } a \in H &\implies a^{-1} \in H \\ &\implies a * a^{-1} \in H \\ &\implies e \in H\end{aligned}$$

- Proof of theorem conti.....

*$\therefore H$ is group under same binary operation
of G .*

$\therefore H$ is a Sub group

Theorem: the necessary and sufficient condition a non empty subset H of a group $(G,*)$ to be a subgroup if $a \in H, b \in H \Rightarrow a * b^{-1} \in H$ where b^{-1} is the Inverse of b in G .

Necessary condition: let H is a subgroup of G .
let $b \in H, b^{-1} \in H$ since H is a group.
 $\Rightarrow a * b^{-1} \in H$

Theorem conti.....

Sufficient Condition: let $a \in H, b \in H \Rightarrow a * b^{-1} \in H$ where b^{-1} is the Inverse of b in G .
We have to show that H is a subgroup.

Now We have to show that H is a subgroup of G .
by the given condition, we have $a \in H, a^{-1} \in H$
 $\Rightarrow a * a^{-1} \in H \Rightarrow e \in H$
 $\therefore$ the identity element exist.

$$a * b^{-1} \in H, \quad \forall a, b \in H, \quad a^{-1}, b^{-1} \in H$$
$$a * a^{-1} = e \in H$$

Theorem conti.....

Again we have $e \in H, a^{-1} \in H$
 $\Rightarrow e * a^{-1} \in H$
 $\Rightarrow a^{-1} \in H$
 $\therefore$ *inverse exist.*

Now, if $b \in H$, then $b^{-1} \in H$
Also $a \in H, b^{-1} \in H \Rightarrow a * (b^{-1})^{-1} \in H$
 $\Rightarrow a * b \in H$
 $\therefore$ *closure properties hold*

Theorem conti.....

Now $H \subseteq G$ and associative law hold good for G ,
as G is a group.

Hence it is true for the elements of H .

$\therefore H$ is a sub group.

Theorem: the intersection of any two subgroup of a group $(G, *)$ is again a subgroup of $(G, *)$.

Proof: let H_1, H_2 be any two subgroup of $(G, *)$ we have $H_1 \cap H_2 \neq \emptyset$

Since at least the identity element is common to both H_1 and H_2

Let $a \in H_1 \cap H_2$ and $b \in H_1 \cap H_2$

$\Rightarrow a \in H_1$ and $a \in H_2$, $b \in H_1$ and $b \in H_2$

$\Rightarrow a, b \in H_1 \Rightarrow a, b^{-1} \in H_1 \Rightarrow \underline{a * b^{-1}} \in H_1$

$\Rightarrow a, b \in H_2 \Rightarrow a, b^{-1} \in H_2 \Rightarrow \underline{a * b^{-1}} \in H_2$

H_1 & H_2 are two subgroups
then $H_1 \cap H_2$ is the subgroup.

Soln :- $H_1 \cap H_2$ is the subgroup.

(i) $H_1 \cap H_2$ is non empty

(ii) st $a, b \in H_1 \cap H_2$ then

$$ab^{-1} \in H_1 \cap H_2$$

then $H_1 \cap H_2$ is the subgroup

(i) $H_1 \cap H_2$ is non empty

$\therefore e \in H_1 \cap H_2$ [$\because$ The identity element of H_1 & H_2 is e]

(ii) $\forall a, b \in H_1 \cap H_2$

$ab^{-1} \in H_1 \cap H_2$

If $a, b \in H_1 \cap H_2 \Rightarrow a, b \in H_1$
 $a, b \in H_2$

$$a, b \in H_1,$$

$\Downarrow$

$$a^{-1}, b^{-1} \in H_1,$$

$\Downarrow$

$$\text{so } a, b^{-1} \in H_1,$$

$$\text{then } ab^{-1} \in H_1,$$

(closure law)

$$\Rightarrow ab^{-1} \in H_1 \cap H_2.$$

$$a, b \in H_2$$

$\Downarrow$

$$a^{-1}, b^{-1} \in H_2$$

$$\text{so } a, b^{-1}$$

$$a, b^{-1} \in H_2$$

$$ab^{-1} \in H_2$$

closure law.

$\Rightarrow 17, nH_2$ is the subgroup

$H_1 \rightarrow \text{subgroup}$
 $H_2 \rightarrow \text{subgroup}$

$\left. \begin{array}{l} H_1 \rightarrow \text{subgroup} \\ H_2 \rightarrow \text{subgroup} \end{array} \right\} H_1 \cap H_2 \text{ is } \underline{\underline{\text{subgroup}}}$

$(1) \quad H_1 \cap H_2 \neq \emptyset$

$\xrightarrow{e} \quad \xrightarrow{e} \quad \text{---}$

$\underline{\underline{\text{subgroup}}}$

$(2) \quad \forall a, b \in H_1 \cap H_2, a \neq b$

$\forall a, b \in H_1 \cap H_2$

$a, b \in H_1 \Rightarrow (ab) \in H_1$
 $a, b^{-1} \in H_1$

$(ab) \in H_2$
 $a, b^{-1} \in H_2$

$\in H_1 \cap H_2$

Proof conti...

$$\therefore a * b^{-1} \in H_1 \cap H_2$$

$\therefore H_1 \cap H_2$ is a subgroup.

Q. Give one example that ,the union of two subgroup is not necessary is a subgroup of G.

Let G be the additive group of integers and H_1, H_2 are the two subgroup of G. that is

$$H_1 = \{\dots -6, -4, -2, 0, 2, 4, 6 \dots\} \text{ and}$$
$$H_2 = \{\dots -9, -6, -3, , 0, 3, 6, 9 \dots\}$$

$$H_1 \cup H_2 = \{\dots -4, -3, -2, 0, \underline{2}, \underline{3}, 4, 6 \dots\}$$

Closer properties not satisfies, since
 $2, 3 \in H_1 \cup H_2$ but $2+3=5 \notin H_1 \cup H_2$

Q. If H_1 and H_2 are two subgroups
 then $\underline{H_1 \cup H_2}$ not necessarily
 a subgroup.

Soln:

$$H_1 \cup H_2$$

$2\mathbb{Z}$

$$H_1 = \{ \dots -3, -2, 0, 2, 3, 4, \dots \}$$

$3\mathbb{Z} \leftarrow$

$$H_2 = \{ \dots -6, -3, 0, 3, 6, 9, 12, \dots \}$$

$$H_1 \cup H_2 = \{ \dots \}$$

$$H_1 \cup H_2 = \{ \dots, 0, 2, 3, 4, \}$$